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Chlorine trifluoride

Chlorine trifluoride is an interhalogen compound with the formula ClF_3. It is a colorless, poisonous, corrosive, and extremely reactive gas that condenses to a pale-greenish yellow liquid, the form in which it is most often sold (pressurized at room temperature). It is notable for its extreme oxidation properties. The compound is primarily of interest in plasmaless cleaning and etching operations in the semiconductor industry,[8][9] in nuclear reactor fuel processing,[10] historically as a component in rocket fuels, and various other industrial operations owing to its corrosive nature.[11]

Preparation, structure, and properties

It was first reported in 1930 by Ruff and Krug who prepared it by fluorination of chlorine; this also produced chlorine monofluoride (ClF) and the mixture was separated by distillation.[12]

$$3\ F_2 + Cl_2 \rightarrow 2\ ClF_3$$

Several hundred tons are produced annually.[13]

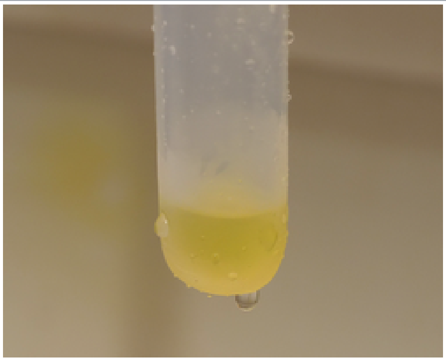
The molecular geometry of ClF_3 is approximately T-shaped, with one short bond (1.598 Å) and two long bonds (1.698 Å).[14] This structure agrees with the prediction of VSEPR theory, which predicts lone pairs of electrons as occupying two equatorial positions of a hypothetic trigonal bipyramid. The elongated Cl-F axial bonds are consistent with hypervalent bonding.

Reactions

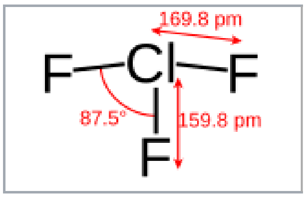
ClF_3 also reacts explosively with water to give hydrogen fluoride and hydrogen chloride, along with oxygen and oxygen difluoride (OF_2):[15]

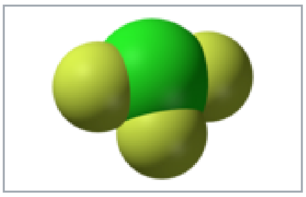
$$ClF_3 + H_2O \rightarrow HF + HCl + OF_2$$

Chlorine trifluoride



Liquid phase Chlorine trifluoride discoloured by the presence of chlorine





Names	
Systematic IUPAC name	Trifluoro-λ³-chlorane[1] *(substitutive)*
Other names	Chlorotrifluoride, CTF
Identifiers	
CAS Number	7790-91-2 (https://commonchemistry.cas.org/detail?cas_rn=7790-91-2) ✓
3D model (JSmol)	Interactive image (https://chemapps.stolaf.edu/jmol/jmol.php?model=F%5BCl%5D%28F%29F)
ChEBI	CHEBI:30123 (https://www.ebi.ac.uk/chebi/searchId.do?chebiId=30123) ✓
ChemSpider	23039 (https://www.chemspider.com/Chemical-Structure.23039.html) ✓
ECHA InfoCard	100.029.301 (https://echa.europa.eu/substance-information/-/substanceinfo/100.029.301)

$ClF_3 + 2H_2O \rightarrow 3HF + HCl + O_2$

Upon heating, it decomposes:[13]

$ClF_3 \rightleftharpoons ClF + F_2$

Reactions with many metals and even metal oxides give fluorides:[15]

$6NiO + 4\ ClF_3 \rightarrow 6\ NiF_2 + 3\ O_2 + 2\ Cl_2$
$AgCl + ClF_3 \rightarrow AgF_2 + ClF + 1/_2\ Cl_2$

ClF_3 is used to produce uranium hexafluoride:

$U + 3\ ClF_3 \rightarrow UF_6 + 3\ ClF$

With phosphorus, it yields phosphorus trichloride (PCl_3) and phosphorus pentafluoride (PF_5), while sulfur yields sulfur dichloride (SCl_2) and sulfur tetrafluoride (SF_4).

It reacts with caesium fluoride to give a salt containing the anion $F(ClF_3)_3^-$.[16]

It reacts with inorganic oxides to give salts of chloryl cation ($[ClO_2]^+$).[17]

Uses

Semiconductor industry

In the semiconductor industry, chlorine trifluoride is used to clean chemical vapour deposition chambers. It can be used to remove semiconductor material from the chamber walls without the need to dismantle the chamber. Unlike most of the alternative chemicals used in this role, it does not need to be activated by the use of plasma since the heat of the chamber is sufficient to make it decompose and react with the semiconductor material.

Fluorination reagent

ClF_3 is used for the fluorination of a variety of compounds.[13]

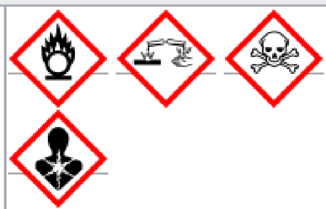
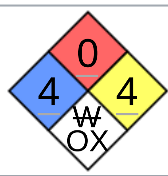
EC Number	232-230-4
Gmelin Reference	1439
MeSH	chlorine+trifluoride (https://www.nlm.nih.gov/cgi/mesh/2014/MB_cgi?mode=&term=chlorine+trifluoride)
PubChem CID	24637 (https://pubchem.ncbi.nlm.nih.gov/compound/24637)
RTECS number	FO2800000
UNII	921841L3N0 (https://precision.fda.gov/uniisearch/srs/unii/921841L3N0) ✓
UN number	1749
CompTox Dashboard (EPA)	DTXSID90893948 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID90893948)
InChI	
InChI=1S/ClF3/c2-1(3)4 ✓ Key: JOHWNGGYGAVMGU-UHFFFAOYSA-N ✓	
InChI=1/ClF3/c2-1(3)4 Key: JOHWNGGYGAVMGU-UHFFFAOYAB	
SMILES	
F[Cl](F)F	
Properties	
Chemical formula	ClF_3
Molar mass	92.45 g·mol^{-1}
Appearance	Colorless gas or greenish-yellow liquid
Odor	Sweet, pungent, irritating, suffocating[2][3]
Density	3.779 g/L[4]
Melting point	−76.34 °C (−105.41 °F; 196.81 K)[4]
Boiling point	11.75 °C (53.15 °F; 284.90 K)[4] (decomposes at 180 °C, 356 °F, 453 K)
Solubility in water	Reacts with water[1]
Solubility	Soluble in carbon tetrachloride but explosive in high concentrations. Reacts with hydrogen-containing

Military applications

Chlorine trifluoride has been investigated as a high-performance storable oxidizer in rocket propellant systems. Handling concerns, however, severely limit its use. The following passage by rocket scientist John D. Clark is widely quoted in descriptions of the substance's extremely hazardous nature:

> It is, of course, extremely toxic, but that's the least of the problem. It is hypergolic with every known fuel, and so rapidly hypergolic that no ignition delay has ever been measured. It is also hypergolic with such things as cloth, wood, and test engineers, not to mention asbestos, sand, and water—with which it reacts explosively. It can be kept in some of the ordinary structural metals—steel, copper, aluminum, etc.—because of the formation of a thin film of insoluble metal fluoride that protects the bulk of the metal, just as the invisible coat of oxide on aluminium keeps it from burning up in the atmosphere. If, however, this coat is melted or scrubbed off, and has no chance to reform, the operator is confronted with the problem of coping with a metal-fluorine fire. For dealing with this situation, I have always recommended a good pair of running shoes.[18]

Under the code name **N-Stoff** ("substance N"), chlorine trifluoride was investigated for military applications by the Kaiser Wilhelm Institute in Nazi Germany not long before the start of World War II. Tests were made against mock-ups of the Maginot Line fortifications, and it was found to be an extremely effective incendiary weapon and poison gas. From 1938, construction commenced on a partly bunkered, partly subterranean 14,000 m^2 (150,000 sq ft) munitions factory, the Falkenhagen industrial complex, which was intended to produce 90 tonnes of N-Stoff per month, in addition to sarin (a deadly nerve agent). However, by the time it was captured by the advancing Red Army in 1945, the factory had produced only

	compounds e.g. hydrogen, methane, benzene, ether, ammonia.[1]
Vapor pressure	175 kPa
Magnetic susceptibility (χ)	-26.5×10^{-6} cm^3/mol[5]
Viscosity	91.82 μPa s
Structure	
Molecular shape	T-shaped molecular geometry
Thermochemistry[6]	
Heat capacity (C)	63.9 J K^{-1} mol^{-1}
Std molar entropy ($S^{\ominus}_{298}$)	281.6 J K^{-1} mol^{-1}
Std enthalpy of formation ($\Delta_f H^{\ominus}_{298}$)	−163.2 kJ mol^{-1}
Gibbs free energy ($\Delta_f G^{\ominus}$)	−123.0 kJ mol^{-1}
Hazards	
Occupational safety and health (OHS/OSH):	
Main hazards	Very toxic, very corrosive, powerful oxidizer, violent hydrolysis[3]
GHS labelling:	
Pictograms	
Signal word	**Danger**
NFPA 704 (fire diamond)	
Flash point	Noncombustible[3]
Lethal dose or concentration (LD, LC):	
LC_{50} (median concentration)	95 ppm (rat, 4 hr) 178 ppm (mouse, 1 hr) 230 ppm (monkey, 1 hr) 299 ppm (rat, 1 hr) [7]
NIOSH (US health exposure limits):	
PEL (Permissible)	C 0.1 ppm (0.4 mg/m^3)[3]

about 30 to 50 tonnes, at a cost of over 100 German Reichsmarks per kilogram.[a] N-Stoff was never used in war.[19][20]

REL (Recommended)	C 0.1 ppm (0.4 mg/m^3)[3]
IDLH (Immediate danger)	20 ppm[3]
Safety data sheet (SDS)	[1] (http://www.ilo.org/dyn/icsc/showcard.display?p_lang=en&p_card_id=0656)
Related compounds	
Related compounds	Chlorine pentafluoride Chlorine monofluoride Bromine trifluoride Iodine trifluoride

Except where otherwise noted, data are given for materials in their standard state (at 25 °C [77 °F], 100 kPa).

✓ verify (what is ✓✗ ?)

Infobox references

Hazards

ClF_3 is a very strong oxidizer. It is extremely reactive with most inorganic and organic materials and will combust many otherwise non-flammable materials without any ignition source. These reactions are often violent and in some cases explosive. Steel, copper, and nickel are not consumed because a passivation layer of metal fluoride will form which prevents further corrosion, but molybdenum, tungsten, and titanium are unsuitable as their fluorides are volatile. ClF_3 will quickly corrode even noble metals like iridium, platinum, or gold, oxidizing them to chlorides and fluorides.

This oxidizing power, surpassing that of oxygen, causes ClF_3 to react vigorously with many other materials often thought of as incombustible and refractory. It ignites sand, asbestos, glass, and even ashes of substances that have already burned in oxygen. In one particular industrial accident, a spill of 900 kg of ClF_3 burned through 30 cm of concrete and 90 cm of gravel beneath.[21][18] There is exactly one known fire control/suppression method capable of dealing with ClF_3—flooding the fire with nitrogen or noble gases such as argon. Barring that, the area must simply be kept cool until the reaction ceases.[22] The compound reacts with water-based suppressors and CO_2, rendering them counterproductive.[23]

ClF_3 causes severe chemical and thermal burns. The products of hydrolysis are mainly hydrofluoric acid and hydrochloric acid, which are usually released as steam or vapor due to the highly exothermic nature of the reaction, and these substances present hazards of their own.

See also

- Chlorine fluoride
- Dioxygen difluoride
- List of stoffs

Explanatory notes

^**a** Using data from Economic History Services[24] and The Inflation Calculator[25] it can be calculated that the sum of 100 Reichsmarks in 1941 is approximately equivalent to US$4,652.50 in 2021. Reichsmark exchange rate values from 1942 to 1944 are fragmentary.

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External links

- National Pollutant Inventory – Fluoride and compounds fact sheet (http://www.npi.gov.au/substances/fluoride-compounds/index.html)
- NIST Standard Reference Database (http://webbook.nist.gov/chemistry/)
- CDC – NIOSH Pocket Guide to Chemical Hazards – Chlorine Trifluoride (https://www.cdc.gov/niosh/npg/npgd0117.html)
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